MATHEMATICS

LIMITS

THEORY SHEET

ENGLISH

MATRIX

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1.1 NOTION OF LIMIT

$f(x) = \dfrac{x^2 - 4}{x - 2} \Rightarrow \qquad f(x) = x + 2 \qquad\qquad x \neq 2$

x	2.01	2.0001	2.00.....1	1.99	1.9999	1.99.....9
y	4.01	4.0001	4.00.....1	3.99	3.9999	3.99.....9

$x \to 2^+ \qquad \| \qquad x \to 2^-$

$y \to 4 \qquad \| \qquad y \to 4$

If $x \to a$, then $y \to l \Rightarrow \lim\limits_{x \to a} f(x) = l$

1.2 LHL, RHL AND EXISTENCE OF LIMIT

The value to which f(x) approaches, as x tends to 'a' from the left hand side ($x \to a^-$) is called left hand limit of f(x at x = a. Symbolically, LHL = $\lim\limits_{x \to a^-} f(x) = \lim\limits_{h \to 0} f(a - h) = f(a^-)$

The value to which f(x) approaches, as x tends to 'a' from the right hand side ($x \to a^+$) is called right hand limit of f(x) at x = a. Symbolically, RHL = $\lim\limits_{x \to a^+} f(x) = \lim\limits_{h \to 0} f(a + h) = f(a^+)$

$\lim\limits_{x \to a} f(x)$ is said to exist as when LHL = RHL = a finite quantity.

Ex.1 $f(x) = \text{sgn}(x)$ at $x = 0$

Ans LHL = -1, RHL = 1

Solve

1. $\lim\limits_{x \to 2} [x]$

Ans. LHL = 1, RHL = 2

2. $\lim\limits_{x \to 2.5} [x]$

Ans. LHL = 2 = RHL

3. $\lim\limits_{x \to 2} [x^2]$

Ans. LHL = 3, RHL = 4

4. $\lim\limits_{x \to 2} [x]^2$

Ans. LHL = 1, RHL = 4

1.3 INDETERMINATE FORMS

$\dfrac{0}{0}, \dfrac{\infty}{\infty}, \infty - \infty, 0 \times \infty, 1^\infty, 0^0, \infty^0$.

Initially we will deal with first five forms only and the other two forms will come up after we have gone through differentiation.

Note : (i)Here 0,1 are not exact, infact both are aproaching to their corresponding values.

Determine whether following is an indeterminate form?

1. $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{\tan x}{\tan 3x}$

Ans. yes

2. $\lim\limits_{x\to\frac{\pi}{2}} \dfrac{\tan x}{\tan 2x}$

Ans. No

3. $\lim\limits_{x\to 0^+} \dfrac{\sin x}{[x]}$

Ans. No

4. $\lim\limits_{x\to 0} x\cot x$

Ans Yes

5. $\lim\limits_{x\to 0^+} [x]\cot x$

Ans. No

6. $\lim\limits_{x\to 0} (\text{cosec } x)^{1/x}$

Ans. No

7. $\lim\limits_{x\to 0} (\sin x)^{\text{cosec } x}$

Ans. No

8. $\lim\limits_{x\to 0} (\sin x)^{\sin x}$

Ans. yes

Remember

$\text{exact } 0 \times \text{Anything} = 0;$ $\qquad \dfrac{\text{exact } 0}{\text{Anything}} = 0;$ $\qquad \dfrac{1}{\text{exact } 0} = \text{Non sense}$

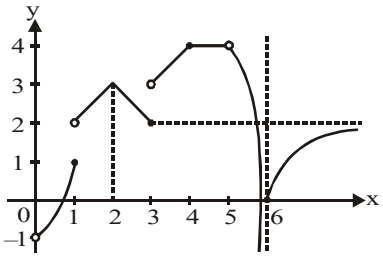
$(\text{exact } 1)^{\text{anything}} = 1;$ $\qquad (\text{anything})^{\text{exact } 0} = 1$ $\qquad (\text{Anything} \neq \text{exact } 0)$

$(\text{exact } 0)^{+\text{ve}} = 0;$ $\qquad (\text{exact } 0)^{\leq 0} = \text{Not defined}$

1.4 READING FROM GRAPHS

Consider the adjacent graph of $y = f(x)$

Find the following :



(a) $\lim_{x\to 0} f(x)$ (b) $\lim_{x\to 0^+} f(x)$ (c) $\lim_{x\to 1^-} f(x)$

(d) $\lim_{x\to 1^+} f(x)$ (e) $\lim_{x\to 2^-} f(x)$ (f) $\lim_{x\to 2^+} f(x)$

(g) $\lim_{x\to 3^-} f(x)$ (h) $\lim_{x\to 3^+} f(x)$ (i) $\lim_{x\to 4^-} f(x)$

(j) $\lim_{x\to 4^+} f(x)$ (k) $\lim_{x\to \infty} f(x)$ (l) $\lim_{x\to 6^-} f(x)$

Ans. (a)–1 (b) –1 (c) 1 (d) 2 (e) 3 (f) 3 (g) 2 (h) 3 (i) 4 (j) 4 (k)2 (l) $-\infty$

Sol. : (a) As $x \to 0^-$: limit does not exist (the function is not defined to the left of $x = 0$)

(b) As $x \to 0^+ : f(x) \to -1 \Rightarrow \lim_{x\to 0^+} f(x) = -1$.(c)As $x \to 1^- : f(x) \to 1 \Rightarrow \lim_{x\to 1^-} f(x) = 1$.

(d) As $x \to 1^+ : f(x) \to 2 \Rightarrow \lim_{x\to 1^+} f(x) = 2$.(e)As $x \to 2^- : f(x) \to 3 \Rightarrow \lim_{x\to 2^-} f(x) = 3$.

(f) As $x \to 2^+ : f(x) \to 3 \Rightarrow \lim_{x\to 2^-} f(x) = 3$.(g)As $x \to 3^- : f(x) \to 2 \Rightarrow \lim_{x\to 3^-} f(x) = 2$.

(h) As $x \to 3^+ : f(x) \to 3 \Rightarrow \lim_{x\to 3^+} f(x) = 3$.(i)As $x \to 4^- : f(x) \to 4 \Rightarrow \lim_{x\to 4^-} f(x) = 4$.

(j) As $x \to 4^+ : f(x) \to 4 \Rightarrow \lim_{x\to 4^+} f(x) = 4$.(k)As $x \to \infty : f(x) \to 2 \Rightarrow \lim_{x\to \infty} f(x) = 2$.

(l) As $x \to 6^-, f(x) \to -\infty \Rightarrow \lim_{x\to 6^-} f(x) = -\infty$ limit does not exist because it is not finite.

1.5 FIVE FUNDAMENTAL THEOREM

Let $\lim_{x\to a} f(x) = \ell$ and $\lim_{x\to a} g(x) = m$. If ℓ & m are finite, then:

(A) $\lim_{x\to a} \{ f(x) \pm g(x) \} = \ell \pm m$

(B) $\lim_{x\to a} \{ f(x).\, g(x) \} = \ell . m$

(C) $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\ell}{m}$, provided $m \neq 0$

(D) $\lim_{x\to a} k f(x) = k \lim_{x\to a} f(x) = k\ell$; where k is a constant.

Type I	Type II
We do not need to calculate LHL and RHL separately	We will calculate LHL and RHL separately for ex. : non uniformly defined function x, [x], {x}, |x|, sgn(x), ITF etc.

2.1 STRATEGIES TO CALCULATE LIMITS

Step - I : Always check indeterminate form first.

Step - II : Simplify, simplify, simplify till the indeterminate form is removed.

2.2 EVALUATING LIMIT BY FACTORIZATION

We can cancel out the factors which are leading to indeterminancy and find the limit of the remaining expression.

Q.1 $\lim\limits_{x\to3} \dfrac{x^2-2x-3}{x^2-4x+3}$

Ans. 2

Sol. : $\lim\limits_{x\to3} \dfrac{x^2-2x-3}{x^2-4x+3} = \lim\limits_{x\to3} \dfrac{(x-3)(x+1)}{(x-3)(x-1)} = 2$

Q.2 $\lim\limits_{x\to 2}\left(\dfrac{2x^2-7x+6}{5x^2-11x+2}\right)$

Ans. 1/9

Sol. $\lim\limits_{x\to 2}\left(\dfrac{2x^2-7x+6}{5x^2-11x+2}\right)$ $\left(\dfrac{0}{0}\text{ form}\right)$

$\Rightarrow \lim\limits_{x\to 2} \dfrac{(x-2)(2x-3)}{(x-2)(5x-1)}$

$\Rightarrow \lim\limits_{x\to 2}\left(\dfrac{2x-3}{5x-1}\right) = \dfrac{4-3}{10-1} = 1/9$

Q.3 $\lim\limits_{x\to1} \dfrac{x^3-3x+2}{x^2-3x+2}$

Ans. 0

Sol. $\lim\limits_{x\to1} \dfrac{(x-1)(x^2+x-2)}{(x-1)(x-2)}$

$\lim\limits_{x\to1} \dfrac{x^2+x-2}{x-2} = 0$

Q.4 $\lim\limits_{x\to 1} \dfrac{x^4-3x+2}{x^5-4x+3}$

Ans. 1

Sol. $\lim\limits_{x\to 1} \dfrac{x^4-3x+2}{x^5-4x+3}$

$$\lim_{x\to 1} \frac{(x-1)\left(x^3+x^3+x-2\right)}{(x-1)\left(x^4+x^3+x^3+x-3\right)} = \frac{1}{1}$$

$= 1$

Q.5 $\lim\limits_{x\to 1} \left(\dfrac{1}{x-1}-\dfrac{3}{x^3-1}\right)$

Ans. 1

Sol. $\lim\limits_{x\to 1}\left(\dfrac{1}{x-1}-\dfrac{3}{x^3-1}\right)$

$$\frac{x^3-3x+2}{(x-1)\left(x^3-1\right)}$$

$$\lim_{x\to 1} \frac{(x-1)^2(x+2)}{(x-1)^2\left(x^2+x+1\right)} = \frac{3}{3} = 1$$

Q.6 $\lim\limits_{x\to 2} \left[\dfrac{1}{x-2}-\dfrac{2(2x-3)}{x^3-3x^2+2x}\right]$

Ans. $-\dfrac{1}{2}$

Sol. : (i) $\lim\limits_{x\to 3} \dfrac{x^2-2x-3}{x^2-4x+3} = \lim\limits_{x\to 3} \dfrac{(x-3)(x+1)}{(x-3)(x-1)} = 2$

(ii) $\lim\limits_{x\to 2} \left[\dfrac{1}{x-2}-\dfrac{2(2x-3)}{x^3-3x^2+2x}\right] = \lim\limits_{x\to 2} \left[\dfrac{1}{x-2}-\dfrac{2(2x-3)}{x(x-1)(x-2)}\right]$

$$= \lim_{x\to 2} \left[\frac{x(x-1)-2(2x-3)}{x(x-1)(x-2)}\right]$$

$$= \lim_{x\to 2}\left[\frac{x^2-5x+6}{x(x-1)(x-2)}\right]$$

$$= \lim_{x\to 2}\left[\frac{(x-2)(x-3)}{x(x-1)(x-2)}\right] = \lim_{x\to 2}\left[\frac{x-3}{x(x-1)}\right] = -\frac{1}{2}$$

Note : In case of $\infty - \infty$ form, change it to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form.

2.3 EVALUATING LIMIT BY RATIONALIZATION

We can rationalise the irrational expression in numerator or denominator or in both to remove the indeterminancy.

$$a-b=\frac{a^2-b^2}{a+b}; \qquad a-b=\frac{a^3-b^3}{a^2+ab+b^2}$$

Q.1 $\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x}$

Ans. 1

Sol. : The form of the given limit is $\frac{0}{0}$ when $x \to 0$. Rationalising the numerator, we get

$$\lim_{x\to 0}\frac{\sqrt{1+x}-\sqrt{1-x}}{x} = \lim_{x\to 0}\left[\frac{\sqrt{1+x}-\sqrt{1-x}}{x}\times\frac{\sqrt{1+x}+\sqrt{1-x}}{\sqrt{1+x}+\sqrt{1-x}}\right]$$

$$= \lim_{x\to 0}\left[\frac{(1+x)-(1-x)}{x\,(\sqrt{1+x}+\sqrt{1-x})}\right] = \lim_{x\to 0}\left[\frac{2x}{x\,(\sqrt{1+x}+\sqrt{1-x})}\right]$$

$$= \lim_{x\to 0}\left[\frac{2}{\sqrt{1+x}+\sqrt{1-x}}\right] = \frac{2}{2} = 1$$

Q.2 $\lim_{x\to 1}\frac{x^{1/3}-1}{x^2+5x-6}$

Ans. $\frac{1}{21}$

Sol. $\lim_{x\to 1}\frac{(x-1)}{(x+6)(x-1)\left(x^{\frac{2}{3}}+x^{\frac{1}{3}}+1\right)} = \frac{1}{21}$

Q.3 $\lim_{x\to 5}\frac{1-\sqrt{x-4}}{x-5}$

Ans. $-\frac{1}{2}$

Sol. $\lim\limits_{x \to 5} \dfrac{1-\sqrt{x-4}}{x-5}$

$\Rightarrow \lim\limits_{x \to 5} \dfrac{1-\sqrt{x-4}}{x-5} \cdot \dfrac{1+\sqrt{x-4}}{1+\sqrt{x-4}}$

$\Rightarrow \lim\limits_{x \to 5} \dfrac{1-x+4}{(x-5)\left(1+\sqrt{x-4}\right)}$

$= \lim\limits_{x \to 5} \dfrac{-(x-5)}{(x-5)\left(1+\sqrt{x-4}\right)}$

$\Rightarrow \lim\limits_{x \to 5} \dfrac{-1}{\left(1+\sqrt{x-4}\right)} = \dfrac{-1}{\left(1+\sqrt{5-4}\right)} = \dfrac{-1}{2}$

Q.4 $\lim\limits_{x \to 0} \dfrac{x}{\sqrt{4+x}-\sqrt{4-x}}$

Ans. 2

Sol. $\lim\limits_{x \to 0} \dfrac{x\left(\sqrt{4+x}+\sqrt{4-x}\right)}{4+x-4+x} = 2$

Q.5 $\lim\limits_{x \to 1} \dfrac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}}$

Ans. $\dfrac{5}{2}$

Sol. : $\lim\limits_{x \to 1} \dfrac{4-\sqrt{15x+1}}{2-\sqrt{3x+1}} = \lim\limits_{x \to 1} \dfrac{(4-\sqrt{15x+1})(2+\sqrt{3x+1})(4+\sqrt{15x+1})}{(2-\sqrt{3x+1})(4+\sqrt{15x+1})(2+\sqrt{3x+1})}$

$= \lim\limits_{x \to 1} \dfrac{(15-15x)}{(3-3x)} \times \dfrac{2+\sqrt{3x+1}}{4+\sqrt{15x+1}} = \dfrac{5}{2}$

Q.6 $\lim\limits_{x \to 1} \dfrac{\sqrt{5-x}-2}{\sqrt{2-x}-1}$

Ans. $\dfrac{1}{2}$

Sol. $\lim\limits_{x \to 1} \dfrac{5-x-4}{2-x-1} \times \dfrac{\sqrt{2-x}+1}{\sqrt{5-x}+2} = \dfrac{1}{2}$

Q.7 If $\lim\limits_{x\to 0}\dfrac{\sqrt{ax+b}-2}{x}=1$ find *a* and *b*.

Ans. $a=4,\ b=4$

Sol. Rationalising, $\lim\limits_{x\to 0}\dfrac{ax+b-4}{x\left(\sqrt{ax+b}+2\right)}$; for existence of limit, $\boxed{b=4}$

so $\lim\limits_{x\to 0}\dfrac{a}{\sqrt{ax+b}+2}=1$; $\dfrac{a}{4}=1 \Rightarrow \boxed{a=4}$

2.4 EVALUATING LIMIT BY SUBSTITUTION

Sometimes in solving limit problem we convert $\lim\limits_{x\to a} f(x)$ into $\lim\limits_{h\to 0} f(a+h)$ or $\lim\limits_{h\to 0} f(a-h)$ according as need of the problem. (here h is approaching to zero.)

Q.1 $\lim\limits_{x\to\frac{\pi}{4}}\dfrac{-1+\cot^3 x}{-2+\cot x+\cot^3 x}$

Ans 3/4

Sol Put $\cot x = t$

$$\lim_{t\to 1}\frac{t^3-1}{t^3+t-2}=\lim_{t\to 1}\frac{(t-1)(t^2+t+1)}{(t-1)(t^2+t+2)}$$

2.5 INFINITY/INFINITY FORM

In these types of problems we usually cancel out the greatest power of x common in numerator and denominator both.

When $x\to-\infty$, we use to substitute $y=-x$ and in this case $y\to\infty$

Also sometime when $x\to\infty$, we use to substitute $y=\dfrac{1}{x}$ and in this case $y\to 0^+$.

Ex.1 Evaluate $\lim\limits_{x\to\infty}\dfrac{x-2}{2x-3}$

$$\lim_{x\to\infty}\frac{a_0x^m+a_1x^{m-1}+.....}{b_0x^n+b_1x^{n-1}+.....}=\begin{cases}\dfrac{a_0}{b_0}, & \text{when } m=n\\ 0, & \text{when } m<n\\ \infty, & \text{when } m>n \text{ and } \dfrac{a_0}{b_0}>0\\ -\infty, & \text{when } m>n \text{ and } \dfrac{a_0}{b_0}<0\end{cases}$$

1. $\lim\limits_{x\to\infty}\dfrac{x^5+7x^2}{x^3-5x^2+2}$

Ans. DNE

2. $\lim\limits_{x\to\infty}\dfrac{\sqrt{x^4-2x^2}}{3x^2-5}$

Ans. $\frac{1}{3}$

Sol. $\lim\limits_{x\to\infty}\dfrac{\sqrt{1-\dfrac{2}{x^2}}}{3-\dfrac{5}{x^2}}=\dfrac{1}{3}$

3. $\lim\limits_{x\to\pm\infty}\left(\sqrt{x^2+1}-x\right)$

Ans. $x\to\infty$ Limit $=0$; $x\to-\infty$ Limit $=$ DNE

Sol. (I) $\lim\limits_{x\to\infty}\left(\dfrac{x^2+1-x^2}{\sqrt{x^2+1}+x}\right)=0$

(II) $\lim\limits_{x\to-\infty}\left(\sqrt{1+x^2}-x\right)$

Let $x=-y$

$\lim\limits_{y\to\infty}\left(\sqrt{1+y^2}+y\right)=$ DNE

4. $\lim\limits_{x\to\infty}\dfrac{|x+3|+|x+1|}{|x+3|-|x+1|}$

Ans. 1

Sol. $\lim\limits_{x\to\infty}\dfrac{(x+1)![(x+3)(x+2)+1]}{(x+1)![(x+3)(x+2)-1]}=1$

5. $\lim\limits_{x\to\pm\infty}\dfrac{\sqrt{2x^2+1}}{5x+3}$

Ans. $x\to\infty,\ \text{Limit}=\dfrac{\sqrt{2}}{5}$; $x\to-\infty,\text{Limit}=-\dfrac{\sqrt{2}}{5}$

Sol. (I) $\lim\limits_{x\to+\infty}\dfrac{x\sqrt{2+\dfrac{1}{x^2}}}{x\left(5+\dfrac{x}{x}\right)}=\dfrac{\sqrt{2}}{5}$

(II) Let $x = -y$

$$\lim_{y\to\infty} \frac{\sqrt{2y^2+1}}{-5y+3} = -\frac{\sqrt{2}}{5}$$

6. $\lim\limits_{x\to\pm\infty} \left(\sqrt{x^2-5x-1} - \sqrt{x^2-3x+1}\right)$

Ans. $x \to \infty,\ \text{Limit} = -1$; $x \to -\infty, \text{Limit} = +1$

Sol. (I) $\lim\limits_{x\to\infty} \dfrac{x^2-5x-1-x^2+3x-1}{\sqrt{x^2-5x-1}+\sqrt{x^2-3x+1}}$

$$= \frac{-2x-2}{x\left(\sqrt{1-\frac{5}{x}-\frac{1}{x^2}}+\sqrt{1-\frac{3}{x}+\frac{1}{x^2}}\right)} = -1$$

(II) Let $x = -y$

$$\lim_{y\to\infty} \sqrt{y^2+5y-1} - \sqrt{y^2+3y+1}$$

$$\lim_{y\to\infty} \frac{2y-2}{\sqrt{y^2+5y-1}+\sqrt{y^2+3y+1}} = +1$$

7. $\lim\limits_{x\to\infty} \left(\sqrt{x+\sqrt{x}} - \sqrt{x}\right)$

Ans. $\dfrac{1}{2}$

Sol. $\lim\limits_{x\to\infty} \dfrac{x+\sqrt{x}-x}{\sqrt{x+\sqrt{x}}+\sqrt{x}} = \dfrac{1}{2}$

8. $\lim\limits_{x\to\infty} \dfrac{5^x+3^x}{5^x-3^x}$

Ans. 1

Sol. $\lim\limits_{x\to\infty}\dfrac{5^x\left(1+\left(\dfrac{3}{5}\right)^x\right)}{5^x\left(1-\left(\dfrac{3}{5}\right)^x\right)}=1$

9. $\lim\limits_{x\to\infty}\dfrac{5^x-3^x}{9^x-2^x}$

Ans. 0

Sol. $\lim\limits_{x\to\infty}\dfrac{9^x\left(\left(\dfrac{5}{9}\right)^x-\left(\dfrac{3}{9}\right)^x\right)}{9^x\left(1-\left(\dfrac{2}{9}\right)^x\right)}=0$

10. $\lim\limits_{n\to\infty}\dfrac{1+2+3.....n}{n^2}$

Ans. $\dfrac{1}{2}$

Sol. $\lim\limits_{n\to\infty}\dfrac{n(n+1)}{2n}=\dfrac{1}{2}$

11. $\lim\limits_{x\to\infty}\dfrac{1^2+2^2+3^2+.....+x^2}{x^3}$

Ans. 1/3

Sol. : $\lim\limits_{x\to\infty}\dfrac{1^2+2^2+3^2+.....+x^2}{x^3}$

$=\lim\limits_{x\to\infty}\dfrac{x(x+1)(2x+1)}{6x^3}$

$=\lim\limits_{x\to\infty}\dfrac{(1+1/x)(2+1/x)}{6}=1/3$

12. $\lim\limits_{n\to\infty}\dfrac{1^3+2^3+3^3.....n^3}{n^4}$

Ans. $\frac{1}{4}$

Sol. $\lim\limits_{n\to\infty}\frac{n^2\left(n^2+1\right)}{4n^4}=\frac{1}{4}$

13. $\lim\limits_{n\to\infty}\ \frac{1^{99}+2^{99}+.....n^{99}}{n^{100}}$

Ans. $\frac{1}{100}$

Sol. $\lim\limits_{n\to\infty}\frac{\left(n+1\right)^{99}}{\left(n+1\right)^{100}-n^{100}}$

$$\Rightarrow \lim_{n\to\infty}\frac{n^{99}\left(1+\frac{1}{n}\right)^{99}}{{}^{100}C_1n^{99}+{}^{100}C_2n^{98}+....}=\frac{1}{100}$$

14. $\lim\limits_{x\to\infty}\ \left(\sqrt{(x+a)(x+b)}-x\right)$

Ans. $\frac{a+b}{2}$

Sol. $\lim\limits_{x\to\infty}\frac{x^2+\left(a+b\right)x+ab-x^2}{\sqrt{x^2+\left(a+b\right)x+ab}+x}=\frac{a+b}{2}$

2.6 STANDARD EXPANSIONS

The following expansions of some standard functions are given-

(i) $e^x = 1 + x + \frac{x^2}{2!}+\frac{x^3}{3!}+.....$

(ii) $\log(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-.....$

(iii) $\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-.....$

(iv) $\cos x =1-\frac{x^2}{2!}+\frac{x^4}{4!}-.....$

(v) $\tan x = x+\frac{x^3}{3}+\frac{2}{15}x^5+.....$

(vi) $\tan^{-1}x=x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+.....$

(vii) $(1+x)^n=1+nx+\frac{n(n-1)}{2!}x^2+..... \ n\in Q$

(viii) $\sin^{-1}x=x+\frac{x^3}{3!}+\frac{9x^5}{5!}.....|x|\le 1$

3.1 APPLICATION OF STANDARD LIMITS TYPE 1

$$\lim_{x\to 0}\frac{(1+x)^n-1}{x}=n$$

If $s\to 1$, then $((s)^n-1)=n(s-1)$

1. $\lim\limits_{x\to 0}\ \dfrac{(1+x)^{1/7}-1}{x^4+4x}$

Ans. $\dfrac{1}{28}$

Sol. $\lim\limits_{x\to 0}\dfrac{\frac{1}{7}(1+x-1)}{x\left(4+x^3\right)}=\lim\limits_{x\to 0}\dfrac{1}{7\left(4+x^3\right)}=\dfrac{1}{28}$

2. $\lim\limits_{x\to 1}\ \dfrac{x^{1/9}-1}{x^2-3x+2}$

Ans. $-\dfrac{1}{9}$

Sol. $\lim\limits_{x\to 1}\dfrac{\frac{1}{9}(x-1)}{(x-1)(x-2)}=-\dfrac{1}{9}$

3. $\lim\limits_{x\to 1}\ \dfrac{x^m-1}{x-1}$

Ans. m

Sol. $\lim\limits_{x\to 1}\dfrac{m(x-1)}{(x-1)}=m$

4. $\lim\limits_{x\to 1}\ \dfrac{x^a-1}{x^b-1}$

Ans. $\dfrac{a}{b}$

Sol. $\lim\limits_{x\to 1}\dfrac{a(x-1)}{b(x-1)}=\dfrac{a}{b}$

5. $\lim\limits_{x\to \pi/2}\dfrac{(1+5\cos x)^{1/6}-1}{2\cos\ x}$

Ans. $\dfrac{5}{12}$

Sol. $\lim\limits_{x\to \frac{\pi}{2}}\dfrac{\frac{1}{6}(1+5\cos x-1)}{2\cos x}=\dfrac{5}{6\times 2}$

6. $\lim\limits_{x\to a}\dfrac{x^n-a^n}{x-a}=na^{n-1}$

Sol. $\lim\limits_{x\to a}\dfrac{a^n\left(\left(\frac{x}{a}\right)^n-1\right)}{a\left(\frac{x}{a}-1\right)}=\lim\limits_{x\to a}\dfrac{na^{n-1}\left(\left(\frac{x}{a}\right)-1\right)}{\left(\frac{x}{a}-1\right)}$

$=na^{n-1}$

7. $\lim\limits_{x\to 1}\dfrac{x^m-x^n}{x-1}$

Ans. $m-n$

Sol. $\lim\limits_{x\to 1}\dfrac{(x^n)(x^{m-n}-1)}{(x-1)}=\lim\limits_{x\to 1}\dfrac{(1)(m-n)(x-1)}{(x-1)}=m-n$

8. $\lim\limits_{x\to 1}\dfrac{x+x^2+x^3-3}{x-1}$

Ans. 6

Sol. $\lim\limits_{x\to 1}\dfrac{(x-1)+(x^2-1)+(x^3-1)}{(x-1)}=\lim\limits_{x\to 1}(1+x+1+x^2+x+1)$

$=6$

9. $\lim\limits_{x\to 1} \dfrac{x^{10}-1}{x^2+x-2}$

Ans. $\dfrac{10}{3}$

Sol. $\lim\limits_{x\to 1}\dfrac{10(x-1)}{(x+2)(x-1)}=\dfrac{10}{3}$

10. $\lim\limits_{x\to 1} \dfrac{(7+x)^{1/3}-2}{x-1}$

Ans. $\dfrac{1}{12}$

Sol. $\lim\limits_{x\to 1}\dfrac{2\left[\left(\dfrac{7+x}{8}\right)^{\frac{1}{3}}-1\right]}{x-1}=\lim\limits_{x\to 1}\dfrac{\frac{2}{3}\left[\dfrac{7+x-8}{8}\right]}{(x-1)}=\dfrac{1}{12}$

11. $\lim\limits_{x\to 1} \dfrac{(7+x)^{1/3}-(3+x)^{1/2}}{x-1}$

Ans. $-\dfrac{1}{6}$

Sol. $\lim\limits_{x\to 1}\dfrac{(7+x)^{\frac{1}{3}}-2}{x-1}-\lim\limits_{x\to 1}\dfrac{(x+3)^{\frac{1}{2}}-2}{x-1}$

$$\frac{1}{12}-\lim_{x\to 1}\frac{2\left[\left(\dfrac{x+3}{4}\right)^{\frac{1}{2}}-1\right]}{x-1}$$

$$\frac{1}{12}-\lim_{x\to 1}\frac{x+3-4}{4(x-1)}=\frac{1}{12}-\frac{1}{4}=-\frac{1}{6}$$

3.2 APPLICATION OF STANDARD LIMITS : TRIGONOMETRIC LIMITS

$$\lim_{x\to 0}\frac{\sin x}{x}=1=\lim_{x\to 0}\frac{\tan x}{x}=\lim_{x\to 0}\frac{\tan^{-1}x}{x}=\lim_{x\to 0}\frac{\sin^{-1}x}{x}$$

$$\text{If } S\to 0 \qquad \left.\begin{matrix}\sin(S)\\ \sin^{-1}(S)\\ \tan(S)\\ \tan^{-1}(S)\end{matrix}\right\}\Rightarrow S$$

1a. $\lim\limits_{x\to 0}\dfrac{\sin 2x}{\sin 3x}$

Ans. $\dfrac{2}{3}$

Sol. : $\lim\limits_{x\to 0}\dfrac{\sin 2x}{\sin 3x}=\lim\limits_{x\to 0}\left[\dfrac{\sin 2x}{2x}\cdot\dfrac{2x}{3x}\cdot\dfrac{3x}{\sin 3x}\right]=\left[\lim\limits_{2x\to 0}\dfrac{\sin 2x}{2x}\right]\cdot\dfrac{2}{3}\cdot\left[\lim\limits_{3x\to 0}\dfrac{3x}{\sin 3x}\right]$

$=1\cdot\dfrac{2}{3}\times\left[\lim\limits_{3x\to 0}\dfrac{\sin 3x}{3x}\right]=\dfrac{2}{3}\times 1=\dfrac{2}{3}$

1b. $\lim\limits_{x\to 0}\dfrac{\sin 2x}{x^4+5x}$

Ans. $\dfrac{2}{5}$

Sol. $\lim\limits_{x\to 0}\dfrac{2x}{x\left(5+x^3\right)}=\dfrac{2}{5}$

2. $\lim\limits_{x\to 0}\dfrac{\tan 4x}{\sin 5x}$

Ans. $\dfrac{4}{5}$

Sol. $\lim\limits_{x\to 0}\dfrac{4x}{5x}=\dfrac{4}{5}$

3. $\lim\limits_{x\to 1}\dfrac{\sin(x-1)}{x^3-1}$

Ans. $\dfrac{1}{3}$

Sol. $\lim\limits_{x\to 1}\dfrac{(x-1)}{(x-1)\left(x^2+x+1\right)}=\dfrac{1}{3}$

4. $\lim\limits_{x\to 0} \dfrac{\sin^2(2x)}{x^2}$

Ans. 4

Sol. $\lim\limits_{x\to 0} \dfrac{(2x)^2}{x^2} = 4$

5. $\lim\limits_{x\to 0} \dfrac{1-\cos x}{x^2} = \dfrac{1}{2}$

Sol. $\lim\limits_{x\to 0} \dfrac{2\sin^2 \dfrac{x}{2}}{x^2} = \dfrac{2\times x^2}{4x^2} = \dfrac{1}{2}$

6. $\lim\limits_{x\to 0} \left(\cot(3x)\right) (\sin 2x)$

Ans. $\dfrac{2}{3}$

Sol. $\lim\limits_{x\to 0} \dfrac{\cos 3x}{\sin 3x} \times \sin 2x = \dfrac{2x}{3x}$

7. $\lim\limits_{x\to 0} \dfrac{\cot 3x}{x\,\text{cosec}^2\, 2x}$

Ans. $\dfrac{4}{3}$

Sol. $\lim\limits_{x\to 0} \dfrac{(\cos 3x)\times \sin^2 2x}{\sin 3x \times x} = \dfrac{(2x)^2}{3x^2} = \dfrac{4}{3}$

8. $\lim\limits_{x\to 1} \dfrac{x^2-4x+3}{\tan(x^2-1)}$

Ans. –1

Sol. $\lim\limits_{x\to 0} \dfrac{(x-1)(x-3)}{(x-1)(x+1)} = -1$

9. $\lim\limits_{x\to 0} \dfrac{1-\cos^4 x}{x\sin 2x}$

Ans. 1

Sol. $\lim\limits_{x\to 0}\dfrac{\left(1-\cos^2 x\right)\left(1+\cos^2 x\right)}{x\times 2x}=\lim\limits_{x\to 0}\dfrac{\left(1-\cos x\right)\left(1+\cos x\right)}{x^2}$

10. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{1-\sin x}{\left(\frac{\pi}{2}-x\right)^2}$

Ans. $\dfrac{1}{2}$

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{1-\cos\left(\frac{\pi}{2}-x\right)}{\left(\frac{\pi}{2}-x\right)^2}=\dfrac{1}{2}$

11. $\lim\limits_{x\to 0}\ \dfrac{\cos x-\cos 9x}{\cos 3x-\cos 7x}$

Ans. 2

Sol. $\lim\limits_{x\to 0}\dfrac{2\sin 5x\cdot\sin 4x}{2\sin 5x\cdot\sin 2x}=2$

12. $\lim\limits_{x\to 0}\ \dfrac{\sin(x^o)}{x}$

Ans. $\dfrac{\pi}{180}$

Sol. $\lim\limits_{x\to 0}\dfrac{\sin\left(\frac{\pi x}{180}\right)}{x}=\dfrac{\pi}{180}$

13. $\lim\limits_{x\to 1}\ (1-x)\tan\left(\dfrac{\pi}{2}x\right)$

Ans. $\dfrac{2}{\pi}$

Sol. Let $x-1=t$

$$\lim_{t\to 0} -t\times\tan\left(\frac{\pi}{2}+\frac{\pi}{2}t\right)$$

$$\lim_{t\to 0}\frac{t\cdot\cos\left(\frac{\pi}{2}t\right)}{\sin\left(\frac{\pi}{2}t\right)}=\frac{2}{\pi}$$

14. $\lim\limits_{x\to 0}\ \dfrac{\cos\left(\dfrac{\pi}{2\cos x}\right)}{x^2}$

Ans. $-\dfrac{\pi}{4}$

Sol. $\lim\limits_{x\to 0}\dfrac{\sin\left(\dfrac{\pi}{2}-\dfrac{\pi}{2\cos x}\right)}{\dfrac{\pi}{2}\left(\dfrac{\cos x-1}{(\cos x)x^2}\right)}=-\dfrac{\pi}{4}$

15. $\lim\limits_{x\to 0}\ \dfrac{1-\cos(1-\cos x)}{\sin^4(x)}$

Ans. $\dfrac{1}{8}$

Sol. $\lim\limits_{x\to 0}\dfrac{1-\cos(1-\cos x)}{(1-\cos x)^2}\times\dfrac{(1-\cos x)^2}{x^4}=\dfrac{1}{8}$

16. $\lim\limits_{x\to\pi/2}\ \dfrac{\tan x}{\tan 3x}$

Ans. 3

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\sin x}{\cos x}\times\dfrac{\cos 3x}{\sin 3x}$

Let $x-\dfrac{\pi}{2}=t$

$$-\lim_{t\to 0}\frac{\cos\left(\frac{3\pi}{2}+3t\right)}{\cos\left(\frac{\pi}{2}+t\right)}=3$$

17. $\lim\limits_{x\to\pi/2}\ \dfrac{\tan 2x}{\tan x}$

Ans. 0

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\dfrac{\sin 2x}{\cos 2x}\times\dfrac{\cos x}{\sin x}=0$

18. $\lim\limits_{x\to 0}\ \dfrac{\tan x-\sin x}{x^3}$

Ans. $\dfrac{1}{2}$

Sol. : $\ell\text{im}\limits_{x\to 0}\ \dfrac{\tan x-\sin x}{x^3}\ =\ \ell\text{im}\limits_{x\to 0}\ \dfrac{\tan x(1-\cos x)}{x^3}\ =\ \ell\text{im}\limits_{x\to 0}\ \dfrac{\tan x\,.\,2\sin^2\frac{x}{2}}{x^3}$

$$=\ \ell\text{im}_{x\to 0}\ \frac{1}{2}\cdot\frac{\tan x}{x}\cdot\left(\frac{\sin\frac{x}{2}}{\frac{x}{2}}\right)^2=\frac{1}{2}\cdot$$

19. $\lim\limits_{x\to 0}\ \dfrac{3\sin x-\sin 3x}{x\,.\tan^2(2x)}$

Ans. 1

Sol. $\lim\limits_{x\to 0}\dfrac{3\sin x-3\sin x+4\sin^3 x}{x\cdot(2x)^2}=\lim\limits_{x\to 0}\dfrac{4x^3}{4x^3}=1$

Ex.1 $\lim\limits_{x\to 0}\sin\left(\dfrac{1}{x}\right)=\text{DNE}$ Ex.2 $\lim\limits_{x\to 0}\cos\left(\dfrac{1}{x}\right)=\text{DNE}$

$$\begin{cases} \lim\limits_{x\to 0} x^p \sin\left(\frac{1}{x}\right) \begin{cases} \longrightarrow 0 & P > 0 \\ \longrightarrow \text{DNE} & P \le 0 \end{cases} \\ \lim\limits_{x\to 0} x^p \cos\left(\frac{1}{x}\right) \begin{cases} \longrightarrow 0 & P > 0 \\ \longrightarrow \text{DNE} & P \le 0 \end{cases} \end{cases}$$

3.3 APPLICATION OF STANDARD LIMITS : EXPONENTIALAND LOGARITHMIC LIMITS

$\lim\limits_{x\to 0} \dfrac{e^x - 1}{x} = 1$. Note: If $s \to 0$ $(e^s - 1) \Rightarrow s$

Ex.1 $\lim\limits_{x \to 0} \dfrac{e^{-x} - 1}{x}$

Ans. -1

Sol. : $\lim\limits_{x \to 0} \dfrac{e^{-x} - 1}{x} = \lim\limits_{y \to 0} \dfrac{e^y - 1}{-y}$ where $-x = y \Rightarrow -\lim\limits_{y \to 0} \dfrac{e^y - 1}{y} = -1$

Ex.2 $\ell\text{im}\limits_{x\to 3} \dfrac{e^x - e^3}{x - 3}$

Ans. e^3

Sol. : Put $y = x - 3$. So, as $x \to 3 \Rightarrow y \to 0$. Thus

$$\ell\text{im}\limits_{x\to 3} \frac{e^x - e^3}{x-3} = \ell\text{im}\limits_{y\to 0} \frac{e^{3+y} - e^3}{y}$$

$$= \ell\text{im}\limits_{y\to 0} \frac{e^3 . e^y - e^3}{y} = e^3 \ \ell\text{im}\limits_{y\to 0} \frac{e^y - 1}{y} = e^3 \ . \ 1 = e^3$$

Ex.3 $\ell\text{im}\limits_{x\to 0} \dfrac{x(e^x - 1)}{1 - \cos x}$

Ans. 2

Sol. $$\ell\text{im}\limits_{x\to 0} \frac{x(e^x-1)}{1-\cos x} = \ell\text{im}\limits_{x\to 0} \frac{x(e^x-1)}{2\sin^2\frac{x}{2}} = \frac{1}{2} \cdot \ell\text{im}\limits_{x\to 0} \left[\frac{e^x - 1}{x} \cdot \frac{x^2}{\sin^2\frac{x}{2}}\right] = 2.$$

Note: If $S \to 0$ $(a^S - 1) = S \cdot \ln a$

Logarithmic Limits : $\lim\limits_{x \to 0} \dfrac{\ell n (1 + x)}{x} = 1$

Note: If $S \to 1$ $(\ln(S)) \Rightarrow (S - 1)$

1. $\lim\limits_{x\to 0} \dfrac{\ln(\cos x)}{\sin^2 x}$

Ans. $\dfrac{-1}{2}$

Sol. $\lim\limits_{x\to 0} \dfrac{\cos x - 1}{x^2} = -\dfrac{1}{2}$

2. $\lim\limits_{x\to 0} \dfrac{e^{\sin 3x} - 1}{\tan 4x}$

Ans. $\dfrac{3}{4}$

Sol. $\lim\limits_{x\to 0} \dfrac{\sin 3x}{4x} = \dfrac{3}{4}$

3. $\lim\limits_{x\to 0} \dfrac{(1+3x)^{2/5} - 1}{e^{5x} - 1}$

Ans. $\dfrac{6}{25}$

Sol. $\lim\limits_{x\to 0} \dfrac{\frac{2}{5}(1+3x-1)}{5x} = \dfrac{6}{25}$

4. $\lim\limits_{x\to \infty} x(e^{\frac{1}{x}} - 1)$

Ans. 1

Sol. $\lim\limits_{x\to \infty} \dfrac{e^{\frac{1}{x}} - 1}{\frac{1}{x}} = \lim\limits_{x\to \infty} \dfrac{\frac{1}{x}}{\frac{1}{x}} = 1$

5. $\lim\limits_{x\to 0} \dfrac{e^{\sin x} - e^x}{\sin x - x}$

Ans. 1

Sol. $\lim\limits_{x\to 0}\dfrac{e^x\left(e^{\sin x-x}-1\right)}{\left(\sin x-x\right)}=\lim\limits_{x\to 0}\dfrac{\left(e^x\right)\left(\sin x-x\right)}{\left(\sin x-x\right)}=1$

6. $\lim\limits_{x\to 0}\ \dfrac{e^{x^2}-e^x}{x}$

Ans. -1

Sol. $\lim\limits_{x\to 0}\dfrac{e^x\left(e^{x^2-x}-1\right)}{x}\Rightarrow\lim\limits_{x\to 0}\dfrac{x\left(x-1\right)}{x}=-1$

7. $\lim\limits_{x\to 0}\ \dfrac{e^{\sin 2x}-e^{\sin x}}{x}$

Ans. 1

Sol. $\lim\limits_{x\to 0}\dfrac{e^{\sin x}\left(e^{\sin 2x-\sin x}-1\right)}{x}=\lim\limits_{x\to 0}\dfrac{e^0\times\left(\sin 2x-\sin x\right)}{x}$

$$\Rightarrow\lim_{x\to 0}\frac{2\cos\frac{3x}{2}\sin\frac{x}{2}}{x}=\lim_{x\to 0}\frac{2\times\frac{x}{2}}{x}=1$$

8. $\lim\limits_{x\to a}\ \dfrac{a^x-a^a}{x-a}$

Ans. $a^a\ln a$

Sol. $\lim\limits_{x\to a}\dfrac{a^a\left(a^{x-a}-1\right)}{x-a}=\lim\limits_{x\to a}\dfrac{a^a\ell na\left(x-a\right)}{\left(x-a\right)}$

9. $\lim\limits_{x\to 0}\ \dfrac{\cos(xe^x)-\cos(xe^{-x})}{x^3}$

Ans. -2

Sol. $\lim\limits_{x\to 0}\dfrac{2\sin\left(\dfrac{xe^x+xe^{-x}}{2}\right)\cdot\sin\left(\dfrac{xe^{-x}-xe^x}{2}\right)}{x^3}$

$$\lim_{x\to 0}\frac{-2\times x^2\left(\frac{e^{2x}+1}{2e^x}\right)\left(\frac{e^{2x}-1}{2e^x}\right)}{x^3}$$

$$-\lim_{x\to 0}\frac{x^2\times 2\left(e^{2x}-1\right)}{2e^0\cdot e^0\cdot x^3}=-\lim_{x\to 0}\frac{2x}{x}=-2$$

10. $\lim\limits_{x\to 0}\ \dfrac{a^x+a^{-x}-2}{x^2}$

Ans. $(\ln a)^2$

Sol. $\lim\limits_{x\to 0}\dfrac{\left(a^x-1\right)^2}{a^x\cdot x^2}=\dfrac{x^2\left(\ell na\right)^2}{x^2}=\left(\ell na\right)^2$

11. $\lim\limits_{x\to 0}\ \dfrac{e^{x^2}-\cos x}{x^2}$

Ans. $\dfrac{3}{2}$

Sol. $\lim\limits_{x\to 0}\left[\dfrac{e^{x^2}-1}{x^2}+\dfrac{1-\cos x}{x^2}\right]$

$$=\left[\frac{x^2}{x^2}+\frac{1}{2}\right]=\frac{3}{2}$$

12. $\lim\limits_{x\to\infty}\ \dfrac{e^{1/x}-1}{2\tan^{-1}(x)-\pi}$

Ans. $-\dfrac{1}{2}$

Sol. $\lim\limits_{x\to\infty}\dfrac{e^{\frac{1}{x}}-1}{-2\left(\frac{\pi}{2}-\tan^{-1}x\right)}$

$$-\frac{1}{2}\lim_{x\to\infty}\frac{\frac{1}{x}}{\cot^{-1}x}\Rightarrow-\frac{1}{2}\lim_{x\to\infty}\frac{\frac{1}{x}}{\tan^{-1}\left(\frac{1}{x}\right)}$$

$$=-\frac{1}{2}$$

13. $\lim\limits_{x\to0}\dfrac{1^x+2^x+3^x+4^x-4}{x}$

Ans. $\ln(24)$

Sol. $\lim\limits_{x\to0}\dfrac{1^x-1+2^x-1+3^x-1+4^x-1}{x}$

$$\Rightarrow\lim_{x\to0}\frac{x\cdot\left(\ell n1+\ell n2+\ell n3+\ell n4\right)}{x}$$

$\Rightarrow \ell n(4!)=\ell n(24)$

14. $\lim\limits_{x\to0}\dfrac{\ln(1+7x)}{e^{2x}-1}$

Ans. $\dfrac{7}{2}$

Sol. $\lim\limits_{x\to0}\dfrac{1+7x-1}{e^{2x}-1}=\lim\dfrac{7x}{2x}=\dfrac{7}{2}$

15. $\lim\limits_{x\to1}\dfrac{\ln x}{e^x-e}$

Ans. $\dfrac{1}{e}$

Sol. $\lim\limits_{x\to0}\dfrac{x-1}{e\left(e^{x-1}-1\right)}=\dfrac{1}{e}\lim\limits_{x\to0}\dfrac{(x-1)}{(x-1)}=\dfrac{1}{e}$

16. $\lim\limits_{x\to0}\dfrac{\ln(1+\sin x)}{3^x-1}$

Ans. $\frac{1}{\ln 3}$

Sol. $\lim\limits_{x\to 0}\frac{\ell n\left(1+\sin x\right)}{3^x-1}=\lim\limits_{x\to 0}\frac{1+\sin x-1}{x\cdot \ell n3}=\frac{1}{\ell n3}$

17. $\lim\limits_{x\to 1}(x-1)\ \log_x 2$

Ans. $\ln 2$

Sol. $\lim\limits_{x\to 1}\frac{(x-1)\cdot \ell n2}{\ell nx}=\ell n2\lim\limits_{x\to 1}\frac{(x-1)}{(x-1)}=\ell n2$

18. $\lim\limits_{x\to 0}\frac{\ln\ (\sec\ x)}{(e^x-1)\sin x}$

Ans. $\frac{1}{2}$

Sol. $\lim\limits_{x\to 0}\frac{-\ell n\left(\cos x\right)}{\left(e^x-1\right)\cdot x}=\lim\limits_{x\to 0}\frac{1-\cos x}{x\times x}=\frac{1}{2}$

19. $\lim\limits_{x\to e}\frac{\ln\ x-1}{x-e}$

Ans. $\frac{1}{e}$

Sol. $\lim\limits_{x\to 0}\frac{\ell nx-\ell ne}{x-e}=\lim\limits_{x\to 0}\frac{\ell n\left(\frac{x}{e}\right)}{e\left(\frac{x}{e}-1\right)}$

$=\frac{1}{e}\lim\limits_{x\to 0}\frac{\frac{x}{e}-1}{\frac{x}{e}-1}=\frac{1}{e}$

20. $\lim\limits_{x\to a}\frac{x^x-a^x}{x-a}$

Ans. a^a

Sol. $\lim\limits_{x\to a}\dfrac{e^{x\ell nx}-e^{x\ell na}}{x-a}=\lim\limits_{x\to a}\dfrac{e^{x\ell na}\left(e^{x\ell nx-x\ell na}-1\right)}{(x-a)}$

$$a^a\lim_{x\to a}\frac{x\left(\ell nx-\ell na\right)}{(x-a)}\Rightarrow a^a\lim_{x\to a}\frac{x\cdot\ell n\left(\frac{x}{a}\right)}{a\left(\frac{x}{a}-1\right)}$$

$$\Rightarrow\frac{a^a\times a}{a}\lim_{x\to a}\frac{\frac{x}{a}-1}{\frac{x}{a}-1}=a^a$$

21. $\lim\limits_{x\to a}\dfrac{x^x-a^a}{x-a}$

Ans. $a^a(1+\ln a)$

Sol. $\lim\limits_{x\to a}\left[\dfrac{x^x-a^x}{x-a}+\dfrac{a^x-a^a}{x-a}\right]$

$=a^a+a^a\,\ell na$

22. $\lim\limits_{x\to a}\dfrac{a^x-x^a}{x-a}$

Ans. $a^a(\ln a-1)$

Sol. $\lim\limits_{x\to a}\left[\dfrac{a^x-a^a}{x-a}-\dfrac{x^a-a^a}{x-a}\right]$

$=a^a\cdot\ell na-a\cdot a^{a-1}$

$=a^a(\ell na-1)$

3.4 ONE RAISE TO INFINITY FORM

If $\lim\limits_{x\to a}f(x)=1$ and $\lim\limits_{x\to a}\phi(x)=\infty$. then, $\lim\limits_{x\to a}f(x)^{\phi(x)}=e^{\lim\limits_{x\to a}(f(x)-1).\phi(x)}$

Note: $\lim\limits_{x\to a}f(g(x))=f\left(\lim\limits_{x\to a}g(x)\right)$ provided f is continuous

1. $\lim\limits_{x\to 0} (1+\sin x)^{1/x}$

Ans. e

Sol. $\lim\limits_{x\to 0}\left(1+\sin x\right)^{\frac{1}{x}}$ (1^{∞})

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{1}{x}(1+\sin x-1)}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{\sin x}{x}} = e^{1}$

2. $\lim\limits_{x\to\infty} \left(\dfrac{x+6}{x+1}\right)^{x+4}$

Ans. e^5

Sol. : As $x\to\infty$; $\lim\limits_{x\to\infty} \dfrac{x+6}{x+1} = 1$ and $(x+4)\to\infty$

$$\lim_{x\to\infty} \left(\frac{x+6}{x+1}\right)^{x+4} = \lim_{x\to\infty} \left[1+\left(\frac{x+6}{x+1}-1\right)\right]^{x+4} = \lim_{x\to\infty} \left[1+\frac{5}{x+1}\right]^{x+4}$$

$$= e^{\lim\limits_{x\to\infty}\left(\frac{5}{x+1}\right)(x+4)} = e^{\lim\limits_{x\to\infty} 5\frac{x+4}{x+1}} = e^5$$

3. $\lim\limits_{x\to\pi\backslash 2} (1+3\cot^2 x)^{\tan^2 x}$

Ans. e^3

Sol. $\lim\limits_{x\to\frac{\pi}{2}}\left(1+3\cot^2 x\right)^{\tan^2 x}$ (1^{∞})

$\Rightarrow e^{\lim\limits_{x\to\frac{\pi}{2}}\left(3\cot^2 x\times\tan^2 x\right)}$ $\Rightarrow e^3$

4. $\lim\limits_{x\to\infty} \left(\dfrac{2x+1}{3x+2}\right)^{x}$

Ans. 0

Sol. $\lim\limits_{x\to\infty}\left(\dfrac{2x+1}{3x+2}\right)^{x}$

$\Rightarrow \left(\dfrac{2}{3}\right)^{\infty} = 0$

5. $\lim\limits_{x\to 0}\left(\dfrac{1+\tan x}{1+\sin x}\right)^{\cos ec^2 x}$

Ans. 1

Sol. $\lim\limits_{x\to 0}\left(\dfrac{1+\tan x}{1+\sin x}\right)^{\frac{1}{\sin^2 x}}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{1}{\sin^2 x}\times\left(\frac{1+\tan x-1-\sin x}{1+\sin x}\right)}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{\tan x(1-\cos x)}{x^2}} \Rightarrow e^0 = 1$

6. $\lim\limits_{x\to 0}\ (\sin x + \cos x)^{1/x}$

Ans. e

Sol. $\lim\limits_{x\to 0}(\sin x+\cos x)^{\frac{1}{x}}$ $\qquad (1^\infty)$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{\sin x+\cos x-1}{x}}$

$\Rightarrow e^{\lim\limits_{x\to 0}\left(\frac{\sin x}{x}+\frac{\cos x-1}{x}\right)} \Rightarrow e^{(1+0)} \Rightarrow e$

7. $\lim\limits_{x\to 0}\left(\dfrac{5}{2+\sqrt{9+x}}\right)^{1/x}$

Ans. $e^{-1/30}$

Sol. $\lim\limits_{x\to 0}\left(\dfrac{5}{2+\sqrt{9+x}}\right)^{\frac{1}{x}}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{1}{x}\left(\frac{5-2-\sqrt{9+x}}{2+\sqrt{9+x}}\right)}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{3-\sqrt{9+x}}{5x}}$

$$\Rightarrow e^{\frac{1}{5}\lim_{x\to 0}\frac{9-9-x}{x\left(3+\sqrt{9+x}\right)}} \Rightarrow e^{-\frac{1}{30}}$$

8. $\lim_{x\to 0} (\cos x)^{\cot^2 x}$

Ans. $e^{-1/2}$

Sol. $\lim_{x\to 0}\left(\cos x\right)^{\cot^2 x}$

$$\Rightarrow e^{\lim_{x\to 0}\cot^2 x\times(\cos x-1)}$$

$$\Rightarrow e^{\lim_{x\to 0}\frac{\cos^2 x}{\sin^2 x}\times(\cos x-1)}$$

$$\Rightarrow e^{\lim_{x\to 0}\frac{\cos x-1}{x^2}} = e^{-\frac{1}{2}}$$

9. $\lim_{x\to 0} (\cos 3x)^{2/x^2}$

Ans. e^{-9}

Sol. $\lim_{x\to 0}\left(\cos 3x\right)^{\frac{2}{x^2}}$ (1^∞)

$$\Rightarrow e^{\lim_{x\to 0}\frac{(\cos 3x-1)}{9x^2}\times 9} \Rightarrow e^{-\frac{1}{9}}$$

10. $\lim_{n\to\infty}\left(\frac{a-1+\sqrt[n]{b}}{a}\right)^n$

Ans. $b^{1/a}$

Sol. $\lim_{n\to\infty}\left(\frac{a-1+b^{\frac{1}{n}}}{a}\right)^n$ (1^∞)

$$\Rightarrow e^{\lim_{n\to\infty} n\left(\frac{a-1+b^{\frac{1}{n}}-a}{a}\right)}$$

$$\Rightarrow e^{\lim_{n\to\infty} n\left(\frac{b^{\frac{1}{n}}-1}{a}\right)} \Rightarrow e^{\lim_{n\to\infty}\frac{n}{a}\times\frac{1}{n}\cdot \ell nb}$$

$$\Rightarrow e^{\ell nb^{\frac{1}{a}}} \Rightarrow b^{\frac{1}{a}}$$

11. $\lim\limits_{x\to 0}\left(\dfrac{1^x+2^x.....+n^x}{n}\right)^{n/x}$

Ans. n!

Sol. $\lim\limits_{x\to 0}\left(\dfrac{1^x+2^x+....+n^x}{n}\right)^{\frac{n}{x}}$ (1^∞)

$$\Rightarrow e^{\lim_{x\to 0}\frac{n}{x}\left(\frac{1^x+2^x+...+n^x-n}{n}\right)}$$

$$\Rightarrow e^{\lim_{x\to 0}\frac{n}{x}\left(\frac{\left(1^x-1\right)+\left(2^x-1\right)+....+\left(n^x-1\right)}{n}\right)}$$

$$\Rightarrow e^{\lim_{x\to 0}\frac{x\left(\ell n1+\ell n2+....+\ell n(n)\right)}{x}} \Rightarrow e^{\ell n(n!)} \Rightarrow n!$$

12. $\lim\limits_{x\to\infty} x^2 \sin\left(\ln\sqrt{\cos\left(\dfrac{\pi}{x}\right)}\right)$

Ans. $-\dfrac{\pi^2}{4}$

Sol. $\lim\limits_{x\to\infty} x^2 \sin\left(\ell n\sqrt{\cos\left(\dfrac{\pi}{x}\right)}\right)$

$$\Rightarrow \lim_{x\to\infty} x^2 \sin\left(\sqrt{\cos\left(\frac{\pi}{x}\right)}-1\right)$$

$$\Rightarrow \lim_{x\to\infty} x^2\left(\sqrt{\cos\left(\frac{\pi}{x}\right)}-1\right) \Rightarrow \lim_{x\to\infty} \frac{\left(\cos\frac{\pi}{x}-1\right)}{\left(\frac{\pi^2}{x^2}\right)} \times \frac{\pi^2}{\left(\sqrt{\cos\frac{\pi}{x}}+1\right)}$$

$$\Rightarrow -\frac{\pi^2}{4}$$

13. $\lim_{x\to\infty}\left(\frac{2\tan^{-1}(x)}{\pi}\right)^x$

Ans. $e^{-\frac{2}{\pi}}$

Sol. $\lim_{x\to\infty}\left(\frac{2\tan^{-1}x}{\pi}\right)^x$

$$\Rightarrow e^{\lim_{x\to\infty} x\left(\frac{2\tan^{-1}x-\pi}{\pi}\right)} \Rightarrow e^{\lim_{x\to\infty} -\frac{2x}{\pi}\left(\frac{\pi}{2}-\tan^{-1}x\right)}$$

$$\Rightarrow e^{\lim_{x\to\infty} \frac{-2x}{\pi}\times\cot^{-1}x} \Rightarrow e^{\lim_{x\to\infty} -\frac{2x}{\pi}\times\tan^{-1}\left(\frac{1}{x}\right)}$$

$$\Rightarrow e^{\lim_{x\to\infty} \frac{-2x}{\pi x}} \Rightarrow e^{-\frac{2}{\pi}}$$

14. $\lim_{x\to0}\left(\tan\left(\frac{\pi}{4}+x\right)\right)^{2/x}$

Ans. e^4

Sol. $\lim_{x\to\infty}\left(\tan\left(\frac{\pi}{4}+x\right)\right)^{\frac{2}{x}}$

$$\Rightarrow e^{\lim_{x\to0}\frac{2}{x}\left(\tan\left(\frac{\pi}{4}+x\right)-1\right)} \Rightarrow e^{\lim_{x\to0}\frac{2}{x}\left[\frac{1+\tan x}{1-\tan x}-1\right]}$$

$$\Rightarrow e^{\lim_{x\to0}\frac{2}{x}\left[\frac{2\tan x}{1-\tan x}\right]} \Rightarrow e^{\lim_{x\to0}\frac{4\tan x}{x}} \Rightarrow e^4$$

15. $\lim_{x\to0}(5^x+3^x-1)^{1/x}$

Ans. 15

Sol. $\lim\limits_{x\to 0}\left(5^x+3^x-1\right)^{\frac{1}{x}}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{5^x+3^x-2}{x}} = e^{\lim\limits_{x\to 0}\frac{5^x-1+3^x-1}{x}} \Rightarrow e^{\lim\limits_{x\to 0}\frac{x(\ell n5+\ell n3)}{x}}$

$\Rightarrow e^{\ell n(15)} = 15$

16. $\lim\limits_{x\to 0}\left(\sin^2\left(\frac{\pi}{2-x}\right)\right)^{\sec^2\left(\frac{\pi}{2-2x}\right)}$

Ans. $e^{-1/4}$

Sol. $\lim\limits_{x\to 0}\left(\sin^2\frac{\pi}{2-x}\right)^{\sec^2\left(\frac{\pi}{2-2x}\right)}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{\sin^2\frac{\pi}{2-x}-1}{\cos^2\left(\frac{\pi}{2-2x}\right)}}$

$\Rightarrow e^{\lim\limits_{x\to 0}\frac{-\cos^2\left(\frac{\pi}{2-x}\right)}{\cos^2\left(\frac{\pi}{2-2x}\right)}}$

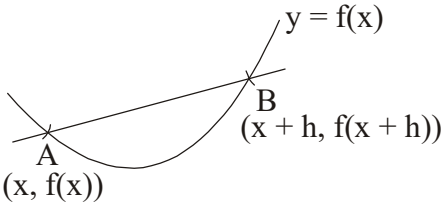
$\Rightarrow e^{\lim\limits_{x\to 0}-\left(\frac{\sin\left(\frac{\pi}{2}-\frac{\pi}{2-x}\right)}{\sin\left(\frac{\pi}{2}-\frac{\pi}{2-2x}\right)}\right)^2}$

$\Rightarrow e^{\lim\limits_{x\to 0}-\left(\frac{\frac{\pi}{2}-\frac{\pi}{2-x}}{\frac{\pi}{2}-\frac{\pi}{2-x}}\right)^2} \Rightarrow e^{\lim\limits_{x\to 0}-\left(\frac{(2-2x)x}{2x(2-x)}\right)^2}$

$\Rightarrow e^{-\left(\frac{1-x}{2-x}\right)^2} \Rightarrow e^{-\frac{1}{4}}$

4. QUESTIONS BASED ON DERIVATIVES

If $B \to A$, line AB will be tangent at A.



$m_{AB} = \frac{f(x+h)-f(x)}{h} \Rightarrow \quad f'(x) = \lim\limits_{h\to 0}\frac{f(x+h)-f(x)}{h}$

Q.1 $\lim\limits_{h\to 0}\dfrac{\tan(x+h)-\tan x}{h}=\dfrac{d(\tan x)}{dx}=\sec^2 x$

$$f''(x)=\lim_{h\to 0}\frac{f(x+2h)-2f(x+h)+f(x)}{h^2}$$

$$f'''(x)=\lim_{h\to 0}\frac{f(x+3h)-3f(x+2h)+3f(x+h)-f(x)}{h^3}$$

$$f''''(x)=\lim_{h\to 0}\frac{f(x+4h)-4f(x+3h)+6f(x+2h)-4f(x+h)+f(x)}{h^4}$$

Relate with

$(h-1)^2=h^2-2h+1$; $(h-1)^3=h^3-3h^2+3h-1$; $(h-1)^4=h^4-4h^3+6h^2-4h+1$

5. SPECIAL TYPE OF FUNCTION

Q.1 $f(x)\begin{cases}x^2 & x\in Q\\ 0 & x\in \bar{Q}\end{cases}$ find (i) $\lim\limits_{x\to 1} f(x)$ (ii) $\lim\limits_{x\to 0} f(x)$

Sol. $x^2=0\Rightarrow$ at $x=0$ limit exists

(i) $\lim\limits_{x\to 1} f(x)=$ DNE

(ii) $\lim\limits_{x\to 0} f(x)=0$

Note : Limit will exist at only those points where both definitions are equal.

6.1 TYPE 2 PROBLEMS: PROBLEMS BASED ON SPECIAL SIGNS

1. $\lim\limits_{x\to 0}\dfrac{\sin x}{|x|}$

Ans. LHL = –1, RHL = 1

Sol. at $x=0$

$$LHL=\lim_{h\to 0}\frac{\sin(0-h)}{|0-h|}=\lim_{h\to 0}-\frac{h}{h}=-1$$

$$RHL=\lim_{h\to 0}\frac{\sin(0+h)}{|0+h|}=\lim_{h\to 0}\frac{h}{h}=1$$

2. $\lim\limits_{x\to 1}\dfrac{\sin(x-1)}{|x|-1}$

Ans. 1

Sol. $\lim\limits_{x\to1}\dfrac{\sin(x-1)}{(x-1)}\Rightarrow\lim\limits_{x\to1}\dfrac{(x-1)}{(x-1)}=1$

3. $\lim\limits_{x\to0}\dfrac{1}{x}\sqrt{\dfrac{1-\cos2x}{1+\cos2x}}$

Ans. LHL = –1, RHL = 1

Sol. $\lim\limits_{x\to0}\dfrac{|\tan x|}{x}$

$$LHL=\lim_{h\to0}\frac{|\tan(0-h)|}{0-h}=-1$$

$$RHL=\lim_{h\to0}\frac{|\tan(0+h)|}{0+h}=1$$

4. $\lim\limits_{x\to0}\dfrac{[x]}{x}$

Ans. LHL = DNE, RHL = 0

Sol. at x = 0

$$LHL=\lim_{h\to0}\frac{[0-h]}{0-h}=DNE$$

$$RHL=\lim_{h\to0}\frac{[0+h]}{h}=0$$

5. $\lim\limits_{x\to5}\dfrac{x^2-9x+20}{x-[x]}$

Ans. LHL = 0, RHL = 1

Sol. $\lim\limits_{x\to5}\dfrac{(x-5)(x-4)}{\{x\}}$

$$LHL=\lim_{h\to0}\frac{(5-h-5)(5-h-4)}{(5-h)-4}=0$$

$$RHL=\lim_{h\to0}\frac{(5+h-5)(5+h-4)}{(5+h)-4}=1$$

6. $\lim\limits_{x\to 0}\ [\sin x]$

Ans. $LHL = -1, RHL = 0$

Sol. $\lim\limits_{x\to 0}\left[\sin x\right]$

$$LHL = \lim_{h\to 0}\left[\sin\left(0-h\right)\right] = -1$$

$$RHL = \lim_{h\to 0}\left[\sin\left(0+h\right)\right] = 0$$

7. $\lim\limits_{x\to 0}\ [\cos x]$

Ans. $LHL = 0 = RHL$

Sol. $\lim\limits_{x\to 0}\left[\cos x\right]$

$$LHL = \lim_{h\to 0}\left[\cos\left(0-h\right)\right] = \left[1^{-}\right] = 0$$

$$RHL = \lim_{h\to 0}\left[\cos\left(0+h\right)\right] = \left[1^{-}\right] = 0$$

8. $\lim\limits_{x\to 0}\ [\tan x]$

Ans. $LHL = -1, RHL = 0$

Sol. $\lim\limits_{x\to 0}\left[\tan x\right]$

$$LHL = \lim_{h\to 0}\left[\tan\left(0-h\right)\right] = \left[0^{-}\right] = -1$$

$$RHL = \lim_{h\to 0}\left[\tan\left(0+h\right)\right] = \left[0^{+}\right] = 0$$

9. $\lim\limits_{x\to 0}\ [\tan^2 x]$

Ans. $LHL = 0 = RHL$

Sol. $\lim\limits_{h\to 0}\left[\tan^2 x\right] = \left[0^{+}\right] = 0$

10. $\lim\limits_{x\to 0}\ \sin^{-1}[\sin x]$

Ans. $LHL = \dfrac{-\pi}{2}, RHL = 0$

Sol. $\lim\limits_{x\to 0}\sin^{-1}\left[\sin x\right]$

$$\text{LHL} = \lim_{h\to 0} \sin^{-1}\left[\sin(0-h)\right] = \lim_{h\to 0} \sin^{-1}\left[0^-\right]$$

$$= \lim_{h\to 0} \sin^{-1}(-1) = -\frac{\pi}{2}$$

$$\text{RHL} = \lim_{h\to 0} \sin^{-1}\left[\sin(0+h)\right] = \lim_{h\to 0} \sin^{-1}\left[0^+\right] = 0$$

11. $\lim\limits_{x\to 0} \sin^{-1}[\cos x]$

Ans. LHL = 0 = RHL

Sol. $\lim\limits_{x\to 0} \sin^{-1}\left[\cos x\right] = \lim\limits_{x\to 0} \sin^{-1}\left[1^-\right] = \lim\limits_{x\to 0} \sin^{-1}(0) = 0$

12. $\lim\limits_{x\to 0} \sin^{-1}(\sec x)$

Ans. DNE

Sol. $\lim\limits_{x\to 0} \sin^{-1}(\sec x) = \lim\limits_{x\to 0} \sin^{-1}(1^+) = \text{DNE}$

13. $\lim\limits_{x\to 1} \dfrac{\sin^{-1}(x) - \frac{\pi}{2}}{x^2-1}$

Ans. DNE

Sol. $\lim\limits_{x\to 1} \dfrac{\sin^{-1}(x) - \frac{\pi}{2}}{x^2-1}$

$$\Rightarrow \lim_{x\to 1} \frac{-\cos^{-1} x}{x^2-1}$$

$$\Rightarrow \lim_{x\to 1} \frac{\sin^{-1}\sqrt{1-x^2}}{1-x^2} \Rightarrow \lim_{x\to 1} \frac{\sqrt{1-x^2}}{1-x^2} = \text{DNE}$$

14. $\lim\limits_{x\to 0} \dfrac{\sin\{x\}}{\{x\}}$

Ans. LHL = sin 1, RHL = 1

Sol. $\lim\limits_{x\to 0} \dfrac{\sin\{x\}}{\{x\}}$

$$LHL = \lim_{h\to 0}\frac{\sin\{0-h\}}{\{0-h\}} = \frac{\sin(-1)}{(-1)} = \sin(1)$$

$$RHL = \lim_{h\to 0}\frac{\sin h}{h} = 1$$

15. $\lim\limits_{x\to 1}\dfrac{e^{\{x\}}-1}{\{x\}}$

Ans. LHL = e – 1, RHL = 1

Sol. $\lim\limits_{x\to 1}\dfrac{e^{\{x\}}-1}{\{x\}}$

$$LHL = \lim_{h\to 0}\frac{e^{\{1-h\}}-1}{\{1-h\}} = e-1$$

$$RHL = \lim_{h\to 0}\frac{e^{\{1+h\}}-1}{\{1+h\}} = \lim_{h\to 0}\frac{e^{h}-1}{h} = 1$$

16. $\lim\limits_{x\to 0}\dfrac{\sin\{x\}}{\tan x}$

Ans. RHL = 1, LHL = DNE

Sol. $\lim\limits_{x\to 0}\dfrac{\sin\{x\}}{\tan x}$

$$LHL = \lim_{h\to 0}\frac{\sin\{0-h\}}{\tan\{0-h\}} = \lim_{h\to 0}\frac{\sin 1}{-h} = DNE$$

$$RHL = \lim_{h\to 0}\frac{\sin\{0+h\}}{\tan\{0+h\}} = \lim_{h\to 0}\frac{\sinh}{\tanh} = 1$$

17. $\lim\limits_{x\to 3}\dfrac{[x^2]-2[x]+3}{[x^2]-4[x]+3}$

Ans. $\dfrac{7}{3}$

Sol. $\lim\limits_{x\to 3}\dfrac{\left[x^2\right]-2[x]+3}{\left[x^2\right]-4[x]+3}$

$$LHL=\lim_{h\to 0}\frac{\left[(3-h)^2\right]-2[3-h]+3}{\left[(3-h)^2\right]-4[3-h]+3}=\frac{8-4+3}{8-8+3}=\frac{7}{3}$$

18. $\lim\limits_{x\to\infty}\left[\dfrac{x+2}{x-1}\right]$

Ans. 1

Sol. $\lim\limits_{x\to\infty}\left[\dfrac{x+2}{x-1}\right]=\left[1^+\right]=1$

19. $\lim\limits_{x\to\infty}\left[\dfrac{x-2}{x+1}\right]$

Ans. 0

Sol. $\lim\limits_{x\to\infty}\left[\dfrac{x-2}{x+1}\right]=\left[1^-\right]=0$

20. Let a be an integer such that $\lim\limits_{x\to 7}\dfrac{18-[1-x]}{[x-3a]}$ exists, where [t] is greatest integer $\le$ t. Then a is equal to.

Ans. – 6

Sol. $\lim\limits_{x\to 7}\dfrac{18-[1-x]}{[x-3a]}$

$\lim\limits_{x\to 7}\dfrac{18-1-[x]}{[x]-3a}$

$\lim\limits_{x\to 7}\dfrac{17-[-x]}{[x]-3a}$

$\text{R.H.L.}=\dfrac{17-(-7)}{6-3a}=\dfrac{24}{6-3a}$

$\text{L.H.L.}=\dfrac{17-(-8)}{7-3a}=\dfrac{25}{7-3a}$

L.H.L. = R.H.L.

$$\frac{24}{6-3a}=\frac{25}{7-3a}$$

$3a = -18$

$a = -6$

6.2 G.I.F. ON STANDARD LIMITS

$\sin x < x < \tan x \quad \Rightarrow \quad$ only for $x > 0$

$$\frac{\sin x}{x}<1\ ;\frac{x}{\sin x}>1\ ;\ \frac{\sin^{-1}x}{x}>1\ ;\ \frac{x}{\sin^{-1}x}<1$$

$$\frac{\tan x}{x}>1\ ;\ \frac{x}{\tan x}<1\ ;\ \frac{\tan^{-1}x}{x}<1\ ;\ \frac{x}{\tan^{-1}x}>1$$

Q.1 $\lim\limits_{x\to0}\left[\frac{\sin x}{x}\right]=\left[1^-\right]=0$

Q.2 $\left[\lim\limits_{x\to0}\frac{\sin x}{x}\right]=[1]=1$

Q.3 $\lim\limits_{x\to0}\left[\frac{4x}{\tan x}\right]=\left[4^-\right]=3$

Q4 $\lim\limits_{x\to0}\left(\left[\frac{\tan^{-1}4x}{x}\right]+\left[\frac{\tan 4x}{2x}\right]+\left[\frac{\sin^{-1}2x}{x}\right]^2+\left[\frac{x}{\tan^{-1}2x}\right]+\left[\frac{\tan^{-1}3x}{x}\right]^2\right)$

Ans. 13

Sol. $\lim\limits_{x\to0}\left(\left[4\cdot\frac{\tan^{-1}4x}{4x}\right]+\left[4\cdot\frac{\tan 4x}{4x}\right]+\left[2\cdot\frac{\sin^{-1}2x}{2x}\right]^2+\left[\frac{1}{2}\cdot\frac{2x}{\tan^{-1}2x}\right]\right)+\left[3\frac{\tan^{-1}3x}{3x}\right]$

$\Rightarrow [4^-] + [4^+] + [2^+]^2 + \left[\frac{1^+}{2}\right] + [3^-]$

$\Rightarrow 3 + 4 + 4 + 0 + 2 = 13$

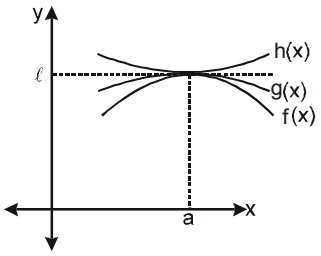
Q.5 $\lim\limits_{x\to0}\left[\frac{e^x-1}{x}\right]$

Sol. $\lim\limits_{x\to0}\left[\frac{e^x-1}{x}\right]=\left[1^+\right]=1=\text{RHL}$

$\text{LHL} = [1^-] = 0$

7. SANDWICH THEOREM

Suppose that $f(x) \le g(x) \le h(x)$ for all x in some open interval containing a, except possibly at x = a itself. Suppose also that



$\lim\limits_{x\to a} f(x) = \ell = \lim\limits_{x\to a} h(x)$,

Then $\lim\limits_{x\to a} g(x) = \ell$.

Note :

1. a can be $\pm\infty$.
2. l can be $\pm\infty$.
3. If $\lim\limits_{x\to a} f(x) \neq \lim\limits_{x\to a} h(x)$, then sandwich theorem is not applicable.

Q.1 $\lim\limits_{x\to 0} x^2 \sin\frac{1}{x} = 0$,

Ans 0

$\because \sin\left(\frac{1}{x}\right)$ lies between –1 & 1

$$\Rightarrow -x^2 \le x^2 \sin\frac{1}{x} \le x^2$$

$$\Rightarrow \lim_{x\to 0} x^2 \sin\frac{1}{x} = 0 \text{ as } \lim_{x\to 0}(-x^2) = \lim_{x\to 0} x^2 = 0$$

Q.2 $\lim\limits_{x\to\infty} \frac{e^{\{x\}}}{x}$

Ans. 0

Sol. $0 \le \{x\} < 1$

$1 \le e^{\{x\}} < e$

$$\frac{1}{x} \le \frac{e^{\{x\}}}{x} < \frac{e}{x}$$

$$\lim_{x\to\infty}\frac{1}{x} \le \lim_{x\to\infty}\frac{e^{\{x\}}}{x} < \lim_{x\to\infty}\frac{e}{x}$$

$\Downarrow_{0}$ $\Downarrow_{0}$ $\Downarrow_{0}$

Q.3 $\lim\limits_{x\to 0}\left(x^2+3x\right)\left[\frac{1}{x}\right]$

Ans. 3

Sol. $\frac{1}{x} - 1 < \left[\frac{1}{x}\right] \le \frac{1}{x}$

$$\frac{\left(x^2+3x\right)(1-x)}{x} < \left(x^2+3x\right)\left[\frac{1}{x}\right] \le \frac{x^2+3x}{x}$$

$$\lim_{x\to 0}(x+3)(1-x) < \lim_{x\to 0}\left(x^2+3x\right)\left[\frac{1}{x}\right] \le \lim_{x\to 0}(x+3)$$

$$\Downarrow_{3} \qquad\qquad \Downarrow_{3} \qquad\qquad \Downarrow_{3}$$

Q.4 $\lim\limits_{n\to\infty}\left(\frac{n}{n^2+1}+\frac{n}{n^2+2}+\frac{n}{n^2+3}+.....+\frac{n}{n^2+n}\right)$

Ans. 1

Sol. $\frac{n}{n^2+n} \le \frac{n}{n^2+1} \le \frac{n}{n^2+1}$

$\frac{n}{n^2+n} \le \frac{n}{n^2+2} \le \frac{n}{n^2+1}$

$\vdots$

add

$\frac{n}{n^2+n} \le \frac{n}{n^2+n} \le \frac{n}{n^2+1}$

$\lim\limits_{n\to\infty}\frac{n^2}{n^2+n} \le \lim\limits_{n\to\infty}\left(\frac{n}{n^2+1}+....+\frac{n}{n^2+n}\right) \le \lim\limits_{n\to\infty}\frac{n^2}{n^2+1}$

1 1 1

Q.5 $\lim\limits_{n\to\infty}\left(\frac{n^2}{n^3+1}+\frac{n^2}{n^3+2}+\frac{n^2}{n^3+3}+.....+\frac{n^2}{n^3+n}\right)$

Ans. 1

Sol. $\lim\limits_{n\to\infty}\frac{n^3}{n^3+n} \le \lim\limits_{n\to\infty}\left(\frac{n^2}{n^3+1}+\frac{n^2}{n^3+2}+....+\frac{n^2}{n^3+n}\right) \le \lim\limits_{n\to\infty}\frac{n^3}{n^3+1}$

1 1 1

Q.6 $\lim\limits_{n\to\infty}\left(\frac{1}{n^2+1}+\frac{2}{n^2+2}+\frac{3}{n^2+3}+.....+\frac{n}{n^2+n}\right)$

Ans. $\frac{1}{2}$

Sol. $\frac{1}{n^2+n} \le \frac{1}{n^2+1} \le \frac{1}{n^2+1}$

$\frac{2}{n^2+n} \le \frac{2}{n^2+2} \le \frac{2}{n^2+1}$

$\vdots$

add

$\frac{n}{n^2+n} \le \frac{n}{n^2+n} \le \frac{n}{n^2+n}$

$\lim_{n\to\infty} \frac{n(n+1)}{2(n^2+n)} \le \lim_{n\to\infty} \left(\frac{1}{n^2+1} + ... + \frac{n}{n^2+n} \right) \le \lim_{n\to\infty} \frac{n(n+1)}{2(n^2+n)}$

$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

Q.7 $\sqrt{5-2x^2} \le f(x) \le \sqrt{5-x^2}$ $\forall\, x \in [-1, 1]$, find $\lim_{x\to 0} f(x)$

Ans. $\sqrt{5}$

Sol. $\lim_{x\to 0} \sqrt{5-2x^2} \le \lim_{x\to 0} f(x) \le \lim_{x\to 0} \sqrt{5-x^2}$

$\sqrt{5}$ $\sqrt{5}$ $\sqrt{5}$

8. LIMITS OF COMPOSITE FUNCTION

Q.1 $f(x) \longrightarrow \begin{cases} |x| & -2 \le x \le 0 \\ e^x - 1 & 0 < x \le 5 \end{cases}$ $g(x) \longrightarrow \begin{cases} -x-1 & -2 \le x \le 0 \\ \frac{4x-3}{3} & 0 < x \le 3 \end{cases}$

Find

(i) $\lim_{x\to 0} f(g(x))$ (ii) $\lim_{x\to 0} f(g(2^x))$ (iii) $\lim_{x\to 0} f([g(2^x - 1)])$, ([.] denotes G.I.F.)

Ans. (i) LHL = 1, RHL = 1 (ii) LHL = $e^{1/3} - 1$ = RHL (iii) LHL = 1 = RHL

Sol. $f(x) = \begin{cases} -x, & -2 \le x \le 0 \\ e^x - 1, & 0 < x \le 5 \end{cases}$, $g(x) = \begin{cases} -x-1, & -2 \le x \le 0 \\ \frac{4x-3}{3}, & 0 < x \le 3 \end{cases}$

(i) $\lim_{x\to 0} f(g(x))$

LHL = f(g(0 – h) = f((–(–h) – 1)) = f(–1 + h) = –(–1) = 1

$$RHL = f(g(0+h)) = f\left(\frac{4h}{3}-1\right) = f(-1) = 1$$

(ii) $\lim\limits_{x\to 0} f\left(g\left(2^{x}\right)\right)$

$$LHL = f\left(g\left(2^{-h}\right)\right) = f\left(\frac{4\cdot 2^{-h}}{3}-1\right) = f\left(\frac{1}{3}\right) = e^{\frac{1}{3}}-1$$

$$RHL = f\left(g\left(2^{h}\right)\right) = f\left(\frac{4\cdot 2^{h}}{3}-1\right) = f\left(\frac{1}{3}\right) = e^{\frac{1}{3}}-1$$

(iii) $\lim\limits_{x\to 0} f\left(\left[g\left(2^{x}-1\right)\right]\right)$

$$LHL = f\left(\left[g\left(2^{-h}-1\right)\right]\right) = f\left[-2^{-h}+1-1\right] = f(-1) = 1$$

$$RHL = f\left[g\left(2^{h}-1\right)\right] = f\left[g(h)\right] = f\left[\frac{4h}{3}-1\right] = f(-1) = 1$$

9.1 LIMITS BASED ON EXPANSION

Ex.1 $\lim\limits_{x\to 0}\dfrac{x^7+3x^3}{x^5+2x^3} = \dfrac{x^3\left(x^4+3\right)}{x^3\left(x^2+2\right)} = \dfrac{3}{2}$

$\Rightarrow$ $\lim\limits_{x\to 0}\dfrac{a_0x^n+a_1x^{n+1}+a_2x^{n+2}+.....}{b_0x^m+b_1x^{m+1}+b_2x^{m+2}+.....} = \dfrac{x^n\left(a_0+a_1x+a_2x^2+....\right)}{x^m\left(b_0+b_1x+b_2x^2+....\right)}$

$n = m$	$n > m$	$n < m$
$\text{limit} = \dfrac{a_0}{b_0}$	$\text{limit} = 0$	$\text{limit} = \text{DNE}$

1. $\lim\limits_{x\to 0}\dfrac{x-\sin x}{x^3}$

Ans. $\dfrac{1}{6}$

Sol. $\lim\limits_{x\to 0}\dfrac{x-\left(x-\dfrac{x^3}{3!}\right)}{x^3} = \dfrac{1}{6}$

2. $\lim\limits_{x\to 0}\dfrac{\tan x-x}{x^3}$

Ans. $\frac{1}{3}$

Sol. $\lim\limits_{x\to 0}\dfrac{x+\dfrac{x^3}{3}-x}{x}=\dfrac{1}{3}$

3. $\lim\limits_{x\to 0}\dfrac{e^x-1-x}{x^2}$

Ans. $\frac{1}{2}$

Sol $\lim\limits_{x\to 0}\dfrac{e^x-1-x}{x^2}=\lim\limits_{x\to 0}\dfrac{\left(1+x+\dfrac{x^2}{2!}+.....\right)-1-x}{x^2}=\dfrac{1}{2}$

4. $\lim\limits_{x\to 0}\dfrac{\ln(1+x)-x}{x^2}$

Ans. $-\frac{1}{2}$

Sol. $\lim\limits_{x\to 0}\dfrac{x-\dfrac{x^2}{2}-x}{x}=-\dfrac{1}{2}$

5. $\lim\limits_{x\to\infty} x-x^2\ln\left(1+\dfrac{1}{x}\right)$

Ans. $\frac{1}{2}$

Sol put $x = 1/y$,

$\lim\limits_{y\to 0}\left[\dfrac{1}{y}-\dfrac{\ell n(1+y)}{y^2}\right]=\dfrac{1}{2}$

6. $\lim\limits_{x\to 0}\dfrac{(1+x)^{1/x}-e}{x}$

Ans. $-\frac{e}{2}$

Sol $= \underset{x\to 0}{\text{Limit}} \dfrac{\left(e^{\overbrace{\frac{\ell n(1+x)}{x}}^{M}-1}-1\right)e}{x} = \underset{x\to 0}{\text{Limit}} \dfrac{e^M-1}{M} \times \underset{x\to 0}{\text{Limit}} \dfrac{\ell n(1+x)-x}{x^2}.e = -\dfrac{e}{2}$

7. $\underset{x\to 0}{\ell im} \dfrac{\ell n(1+x)-\sin x+\frac{x^2}{2}}{x\tan x \sin x}$

Ans. $\dfrac{1}{2}$

Sol. : $\underset{x\to 0}{\ell im} \dfrac{\ell n(1+x)-\sin x+\frac{x^2}{2}}{x\tan x \sin x} = \underset{x\to 0}{\ell im} \dfrac{\left(x-\frac{x^2}{2}+\frac{x^3}{3}-.....\right)-\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-.....\right)+\frac{x^2}{2}}{x^3 . \frac{\tan x}{x} . \frac{\sin x}{x}} = \dfrac{1}{3} + \dfrac{1}{6} = \dfrac{1}{2}$

8. $\underset{x\to 0}{\lim} \dfrac{e^x+\log\{(1-x)/e\}}{\tan x - x}$

Ans. $-1/2$

Sol. $\text{Limit} = \underset{x\to 0}{\lim} \dfrac{e^x+\log(1-x)-1}{\tan x - x}$

$$\underset{x\to 0}{\lim} \dfrac{\left(1+x+\frac{x^2}{2!}+....\right)-x-\frac{x^2}{2}-\frac{x^3}{3}.....-1}{x+\frac{x^3}{3}+\frac{2x^5}{15}+.....-x}$$

$= -1/2$

9. $\underset{x\to 0}{\lim}\left[\dfrac{\sin x \sin^{-1}x - x^2}{x^6}\right]$

Ans. $\dfrac{1}{18}$

Sol. $\underset{x\to 0}{\lim} \dfrac{\left(x-\frac{x^3}{3!}+\frac{x^5}{5!}-...\right)\left(x+\frac{x^3}{3!}+\frac{9x^5}{5!}+....\right)-x^2}{x^6}$

$= \underset{x\to 0}{\lim}$

$$\dfrac{x^2+x^4\left(\frac{1}{3!}-\frac{1}{3!}\right)+x^6\left(\frac{1}{5!}-\frac{1}{3!.3!}+\frac{9}{5!}\right)+x^8(...)+...-x^2}{x^6}$$

$$= \underset{x\to 0}{\lim} \dfrac{x^6\left\{\left[\frac{10}{5!}-\frac{1}{3!.3!}\right]+x^2(...)+...\right\}}{x^6}$$

$$= \frac{10}{120} - \frac{1}{36} = \frac{1}{18}$$

10. $\lim\limits_{x\to 0} \dfrac{e^x - e^{-x} - 2x}{x - \sin x}$

Ans. 2

Sol. : $\lim\limits_{x\to 0} \dfrac{e^x - e^{-x} - 2x}{x - \sin x} \Rightarrow \lim\limits_{x\to 0} \dfrac{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ... - \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + ...\right) - 2x}{x - \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} .....\right)}$

$$\Rightarrow \lim_{x\to 0} \frac{2.\frac{x^3}{6} + 2.\frac{x^5}{5!} + ...}{\frac{x^3}{6} + \frac{x^5}{5!} .....} \Rightarrow \lim_{x\to 0} \frac{x^3\left(\frac{1}{3} + \frac{1}{60}x^2 + ...\right)}{x^3\left(\frac{1}{6} + \frac{1}{120}x^2 + ...\right)} = \frac{1/3}{1/6} = 2$$

9.2 FINDING UNKNOWN CONSTANTS

1. $\lim\limits_{x\to 0} \dfrac{e^x - a - 2bx - 3cx^2}{x^n} = L$ If L exists, find a,b,c & L

(i) $n = 2$ (ii) $n = 3$

Ans. (i) $a = 1, b = \dfrac{1}{2}, C \in R,\ L = \dfrac{1}{2} - 3C$

(ii) $a = 1, b = \dfrac{1}{2}, C = \dfrac{1}{6}\quad L = \dfrac{1}{6}$

Sol. (i) $n = 2$

$$\lim_{x\to 0} \frac{1 + x + \frac{x^2}{2!} - 2bx - 3cx^2 - a}{x^2}$$

$$\frac{(1-a) + (1-2b)x + \left(\frac{1}{2} - 3c\right)x^2}{x^2}$$

$$a = 1, b = \frac{1}{2}, L = \frac{1}{2} - 3c, c \in R$$

(ii) $n = 3$

$$\frac{1-a+(1-2b)x+\left(\frac{1}{2}-3c\right)x^2+\frac{x^3}{3!}}{x^3}$$

$$a=1, b=\frac{1}{2}, c=\frac{1}{6}, L=\frac{1}{6}$$

2. $\lim\limits_{x\to 0}\dfrac{x(1+a\ \cos x)-\sin x}{x^3}=L$; If L exists, find a & L

Ans. $a=0,\ L=\dfrac{1}{6}$

Sol. $$\frac{x+ax\left(1-\frac{x^2}{2!}\right)-x+\frac{x^3}{3!}}{x^3}=L$$

$$\frac{ax+\left(\frac{1}{6}-\frac{a}{2}\right)x^3}{x^3}=L$$

$$a=0, L=\frac{1}{6}$$

3. $\lim\limits_{x\to 0}\dfrac{a\cos x+bx\sin x-5}{x^4}=L$; If L exists, find a,b & L

Ans. $a=5,\ b=\dfrac{5}{2},\ L=\dfrac{-5}{24}$

Sol. $$\frac{a\left(1-\frac{x^2}{2!}+\frac{x^4}{4!}\right)+bx\left(x-\frac{x^3}{3!}\right)-5}{x^4}=L$$

$$\frac{(a-5)+\left(b-\frac{a}{2}\right)x^2+\left(\frac{a}{24}-\frac{b}{6}\right)x^4}{x^4}$$

$$a = 5, b = \frac{5}{2}, L = \frac{5}{24} - \frac{5}{12} = -\frac{5}{24}$$

4. If $\lim\limits_{x\to 0} \dfrac{ae^x - b\cos x + ce^{-x}}{x\sin x} = 2$, then a + b + c is equal to _____.

Ans. 4

Sol. $\lim\limits_{x\to 0} \dfrac{ae^x + ce^{-x} + b\cos x}{x\sin x} = 2$

$$\lim_{x\to 0} \frac{a\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!}.....\right) + c\left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!}.....\right) - b\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!}.....\right)}{x^2} = 2$$

$\Rightarrow$ Coefficient of $x^0 = 0$

$a + c - b = 0$

$\Rightarrow$ Coefficient of $x^1 = 0$

$a - c = 0$

$\Rightarrow$ Coefficient of $x^2 = 2$

$$\frac{a}{2} + \frac{c}{2} + \frac{b}{2} = 2$$

$a + b + c = 4$

10.1 MISCELLANEOUS CONCEPTS: FUNCTIONS WITH BUILT-IN LIMIT

1. Define $f(x) = \lim\limits_{n\to\infty} \dfrac{x^n}{x^n + 1}$, $; x > 0$

Ans. $f(x) = \begin{cases} 1 & x > 1 \\ 1/2 & x = 1 \\ 0 & 0 < x < 1 \end{cases}$

2. Define $f(x) = \lim\limits_{n\to\infty} \dfrac{(1+x)^n \sin x + x}{1 + (1+x)^n}$; $x > -1$

Ans. $f(x) = \begin{cases} \sin x & x > 0 \\ 0 & x = 0 \\ x & -1 < x < 0 \end{cases}$

3. Define $f(x)=\lim_{n\to\infty}\dfrac{e^{nx}(x+1)+(x^2+1)}{e^{nx}+(x+2)}$

Ans. $f(x)=\begin{cases} x+1 & x>0 \\ 2/3 & x=0 \\ \dfrac{x^2+1}{x+2} & x<0 \end{cases}$

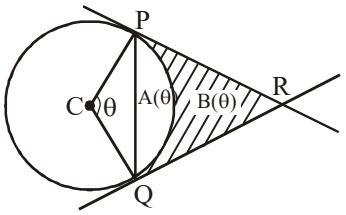
4. Define $f(x)=\lim_{n\to\infty}\dfrac{\tan(\pi x^2)+x^n\sin(\pi x)}{(1+x)^2+x^{n+1}}$; $x>0$

Ans. $f(x)=\begin{cases} \dfrac{\sin(\pi x)}{x} & x>1 \\ 0 & x=1 \\ \dfrac{\tan(\pi x^2)}{(1+x)^2} & 0<x<1 \end{cases}$

10.2 MISCELLANEOUS CONCEPTS: GEOMETRICAL LIMITS

Q.1 An arc PQ of a circle subtends a central angle θ as shown. Let A(θ) be the area between the chord PQ and the arc PQ. Let B(θ) be the area between the tangent lines PR and QR and the arc PQ.

Find $\lim_{\theta\to0}\dfrac{A(\theta)}{B(\theta)}$



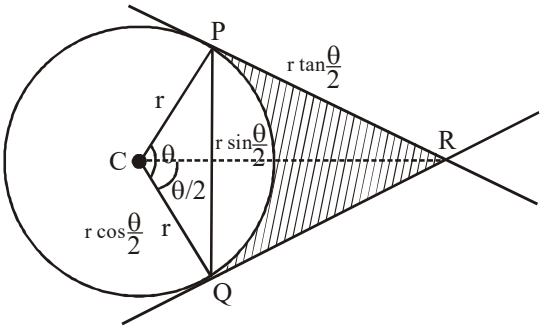
Ans 2

[Sol. $A(\theta)$ = sector CPQ – ΔCPQ

$=\dfrac{1}{2}r^2\theta-\dfrac{1}{2}\cdot r\cdot r\sin\theta$

$=\dfrac{1}{2}r^2\theta-\dfrac{1}{2}r^2\sin\theta$

$=\dfrac{1}{2}r^2(\theta-\sin\theta)$



$B(\theta)$ = Area of quadrilateral PCQR – sector PCQ

$=r^2\tan\dfrac{\theta}{2}-\dfrac{1}{2}r^2\theta=\dfrac{1}{2}r^2\left(2\tan\dfrac{\theta}{2}-\theta\right)$

$\therefore\quad \dfrac{A(\theta)}{B(\theta)}=\dfrac{\theta-\sin\theta}{2\tan(\theta/2)-\theta}=\dfrac{1}{6}\cdot\dfrac{\theta^3}{2\tan(\theta/2)-\theta}$ (use expansion)

$\dfrac{A(\theta)}{B(\theta)}=2$ Ans.